

CSA 0378-DATA STRUCTURES

16. PRIORITY QUEUE – Max heap & Min heap

a.) MAX HEAP

```
main.c
1 #include <stdio.h>
2
3 void heapify(int arr[], int n, int i) {
4     int largest = i;
5     int left = 2 * i + 1;
6     int right = 2 * i + 2;
7
8     if(left < n && arr[left] > arr[largest])
9         largest = left;
10
11    if(right < n && arr[right] > arr[largest])
12        largest = right;
13
14    if(largest != i) {
15        int temp = arr[i];
16        arr[i] = arr[largest];
17        arr[largest] = temp;
18
19        heapify(arr, n, largest);
20    }
21 }
22
23 int main() {
24     int n, i;
25
26     printf("Enter number of elements: ");
27     scanf("%d", &n);
28
29     int arr[n];
30
31     printf("Enter elements: ");
32     for(i = 0; i < n; i++)
33         scanf("%d", &arr[i]);
34
35     for(i = n / 2 - 1; i >= 0; i--)
36         heapify(arr, n, i);
```

Output

```
Enter number of elements: 4
Enter elements: 10 20 30 40
Max Heap: 40 20 30 10
Highest Priority Element: 40

=== Code Execution Successful ===
```

```
main.c
10
11 if(right < n && arr[right] > arr[largest])
12     largest = right;
13
14 if(largest != i) {
15     int temp = arr[i];
16     arr[i] = arr[largest];
17     arr[largest] = temp;
18
19     heapify(arr, n, largest);
20 }
21 }
22
23 int main() {
24     int n, i;
25
26     printf("Enter number of elements: ");
27     scanf("%d", &n);
28
29     int arr[n];
30
31     printf("Enter elements: ");
32     for(i = 0; i < n; i++)
33         scanf("%d", &arr[i]);
34
35     for(i = n / 2 - 1; i >= 0; i--)
36         heapify(arr, n, i);
37
38     printf("Max Heap: ");
39     for(i = 0; i < n; i++)
40         printf("%d ", arr[i]);
41
42     printf("\nHighest Priority Element: %d\n", arr[0]);
43
44     return 0;
45 }
```

Output

```
Enter number of elements: 4
Enter elements: 10 20 30 40
Max Heap: 40 20 30 10
Highest Priority Element: 40

=== Code Execution Successful ===
```

b.) MIN HEAP

main.c

Share

Run

```
1 #include <stdio.h>
2
3 void heapify(int arr[], int n, int i) {
4     int smallest = i;
5     int left = 2 * i + 1;
6     int right = 2 * i + 2;
7
8     if(left < n && arr[left] < arr[smallest])
9         smallest = left;
10
11     if(right < n && arr[right] < arr[smallest])
12         smallest = right;
13
14     if(smallest != i) {
15         int temp = arr[i];
16         arr[i] = arr[smallest];
17         arr[smallest] = temp;
18
19         heapify(arr, n, smallest);
20     }
21 }
22
23 int main() {
24     int n, i;
25
26     printf("Enter number of elements: ");
27     scanf("%d", &n);
28
29     int arr[n];
30
31     printf("Enter elements: ");
32     for(i = 0; i < n; i++)
33         scanf("%d", &arr[i]);
34
35     for(i = n / 2 - 1; i >= 0; i--)
36         heapify(arr, n, i);
```

Output

Enter number of elements: 4
Enter elements: 40 80 102 20
Min Heap: 20 40 102 80
Highest Priority Element: 20

=== Code Execution Successful ===

main.c

Share

Run

```
10
11     if(right < n && arr[right] < arr[smallest])
12         smallest = right;
13
14     if(smallest != i) {
15         int temp = arr[i];
16         arr[i] = arr[smallest];
17         arr[smallest] = temp;
18
19         heapify(arr, n, smallest);
20     }
21 }
22
23 int main() {
24     int n, i;
25
26     printf("Enter number of elements: ");
27     scanf("%d", &n);
28
29     int arr[n];
30
31     printf("Enter elements: ");
32     for(i = 0; i < n; i++)
33         scanf("%d", &arr[i]);
34
35     for(i = n / 2 - 1; i >= 0; i--)
36         heapify(arr, n, i);
37
38     printf("Min Heap: ");
39     for(i = 0; i < n; i++)
40         printf("%d ", arr[i]);
41
42     printf("\nHighest Priority Element: %d\n", arr[0]);
43
44     return 0;
45 }
```

Output

Enter number of elements: 4
Enter elements: 40 80 102 20
Min Heap: 20 40 102 80
Highest Priority Element: 20

=== Code Execution Successful ===